Data-Driven Insights for Strategic Pricing in the Used Car Market

Project Report

1) Introduction:

The used car market has become one of the most dynamic sectors in the automobile industry, driven by the growing affordability gap between new and pre-owned vehicles and the rapid digitalisation of car trading platforms (Dogru, Mody and Suess, 2020). Estimating the fair resale price of a used car has therefore become essential for both individual sellers and dealerships seeking to remain competitive. However, car valuation is inherently complex because resale prices are influenced by a combination of **physical**, **technical**, and **market-based** attributes (Benítez-Aurioles, 2021).

This study investigates the determinants of used car prices using a structured regression approach that progressively incorporates variables such as **age**, **kilometers driven**, **fuel type**, **transmission**, **seller type**, **brand**, and **model**. By developing seven stepwise linear regression models, the analysis aims to examine how each factor individually and collectively contributes to explaining price variation. Exploratory Data Analysis (EDA) methods — like histograms, scatterplots, boxplots, and correlation heatmaps — provide a visual way to examine data patterns prior to fitting models. The findings should provide insights grounded in evidence about the effect of depreciation and brand perception on resale value. These generated relationships will also help users including buyers, sellers, and auto platforms, make data driven pricing decisions that reflect the current state of the market.

2) Problem Identification:

The key business problem identified in this study is the uncertainty in determining fair and accurate resale prices for used cars in a highly competitive and information-asymmetric market.

Buyers and sellers often face difficulties in assessing a vehicle's true market value due to multiple interacting factors.

3) Literature Review:

Numerous studies have investigated the determinants of used car price, and have consistently identified age and mileage, as the main predictors of depreciation (Lee, 2021; Ahamed and Jha, 2023). Automobiles depreciate quickly, in the first few years of life, depreciation is burdensome, and increases as kilometres are driven, as there are more kilometres driven, depreciation is a function of wear and reliability (Kumari and Verma, 2022).

The type of fuel also matters when setting price preference and levels. Diesel vehicles typically, have historically been priced higher based on fuel economy, but recently this trend has shifted more towards petrol and hybrid vehicles based largely on environmental factors (Chattopadhyay, Singh and Mehta, 2022). Transmission type also effects resale value; automatic vehicles are usually sold for higher prices, largely based on driving ease and consumer demand, however this differs by market context (Gupta, 2021).

Factors such as brand reputation and model popularity greatly influence valuation differences. For instance, premium and reliable brands such as Toyota, Honda, and BMW often depreciate more slowly than the lesser brands (Kumar and Singh, 2020). This results from perceived quality, maintenance cost and after-sales service. The type of seller, whether an individual owner or dealer, can also play a role in establishing trust with buyers, which can have an impact on pricing results (Rahman and Alam, 2021).

In general, the literature supports that vehicle depreciation is multifactorial, with measurable factors (e.g. age, mileage) and perceptual factors (e.g. brand image, seller trustworthiness) impacting market price (Benítez-Aurioles, 2021). This research builds on prior work because it models these predictors in stepwise fashion, thus contributing to a deeper understanding of the impact of each set of factors on model fit to the data and overall predictive ability.

While prior studies explored individual determinants, this study uniquely models the combined effects of technical, economic, and perceptual variables stepwise to achieve higher predictive accuracy.

4) Methodology:

Data Source and Preparation

This dataset was sourced from a vehicle resale site and included observations such as selling price, age, kilometers driven, fuel type, transmission type, seller type, make, and model. The dataset was then prepared for modelling by eliminating unnecessary variables, duplicates, and missing values, maintaining only usable records. The names of column labels were standardized for readability, and categorical variables were properly encoded as necessary for regression analysis.

To account for data quality, variables such as price were converted into numeric form and transformed using logarithms to reduce right-skewness and heteroscedasticity. A new variable, **age**, was derived from the manufacturing year to capture depreciation effects, while log (km_driven + 1) was used to represent usage intensity in a more linear form. The dataset was then split into training and testing subsets to evaluate the predictive accuracy of regression models on unseen data.

5) Exploratory Data Analysis (EDA):

To analyze data trends and possible interactions among the variables, exploratory analysis are done prior to building models.

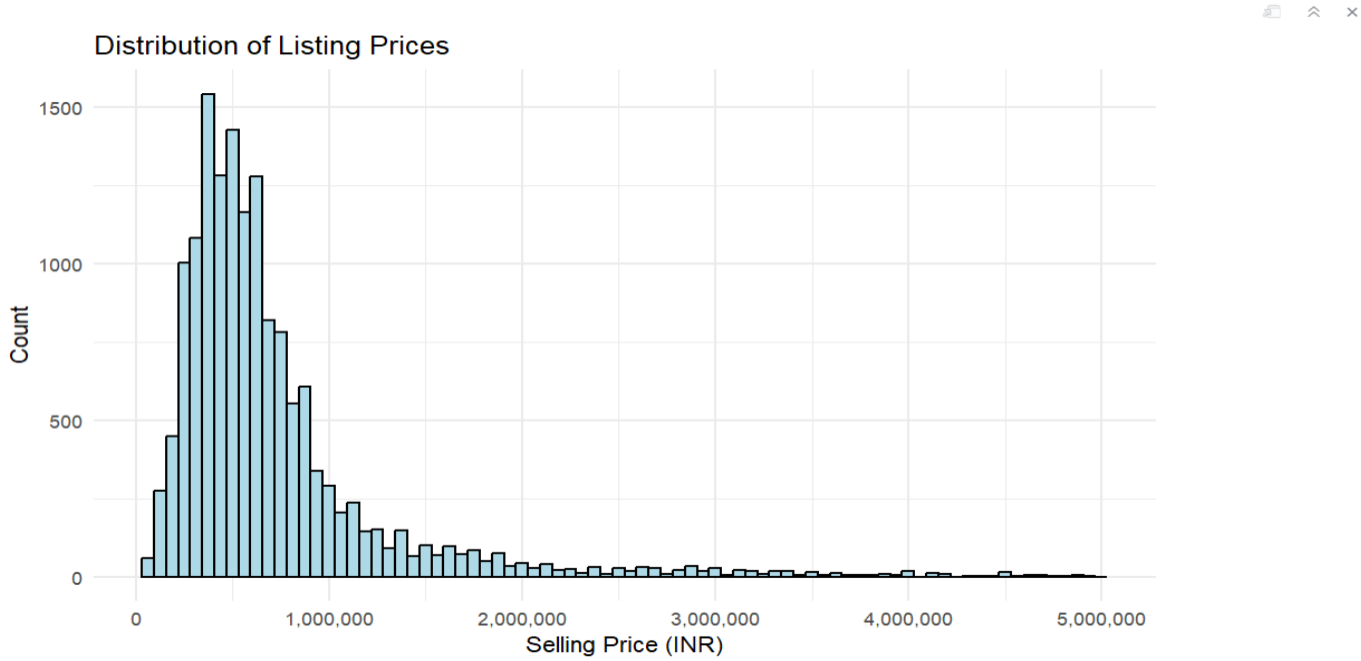

Figure: 1

Figure 1 shows selling price is heavily right-skewed: most vehicles are concentrated in the lower to mid-price bands, while a small number of high-end listings produce a long right tail. Because of this distributional shape we model log(price) to stabilise variance and ease interpretation.

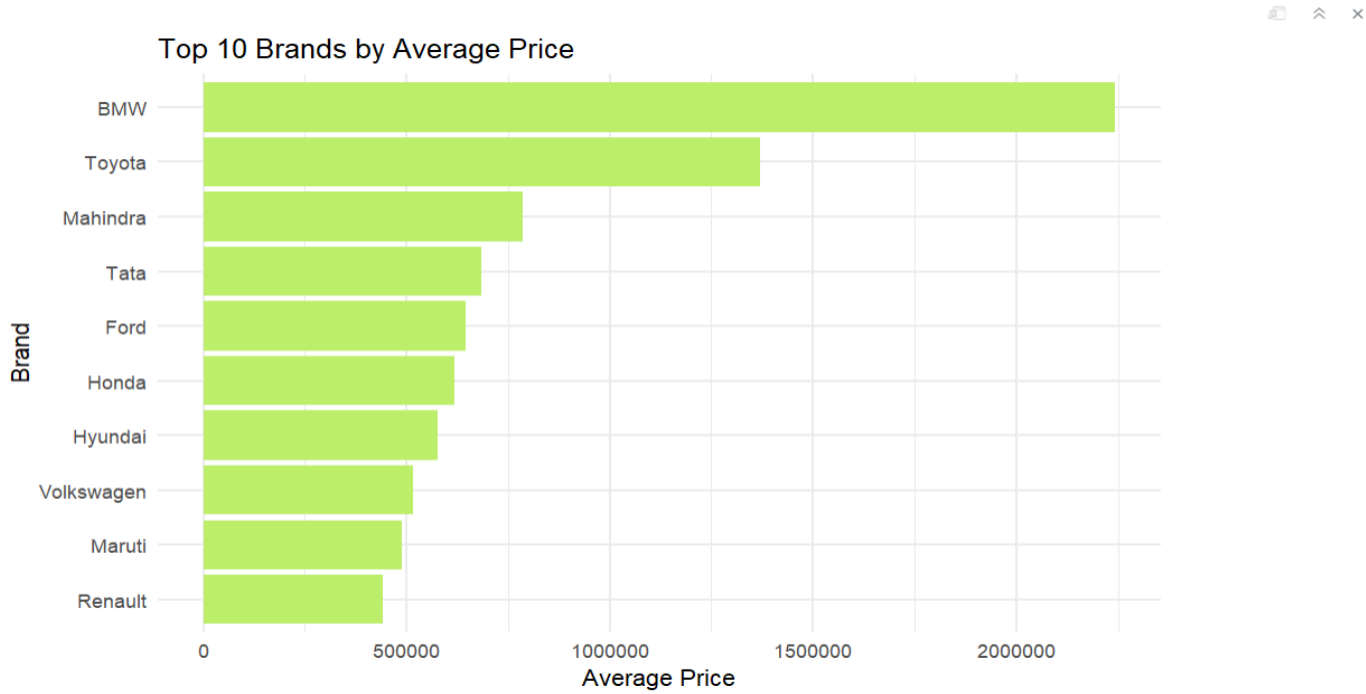

Figure: 2

Figure 2 depicts evident brand effects: a small number of brands have higher mean resale prices -- a statistically prominent average suggesting that brand matters for used-car resale value. Therefore, the brand dummy variables are appropriate to include as predictors in future regressions, but one must think about how to include relatively few categories and stay within model names.

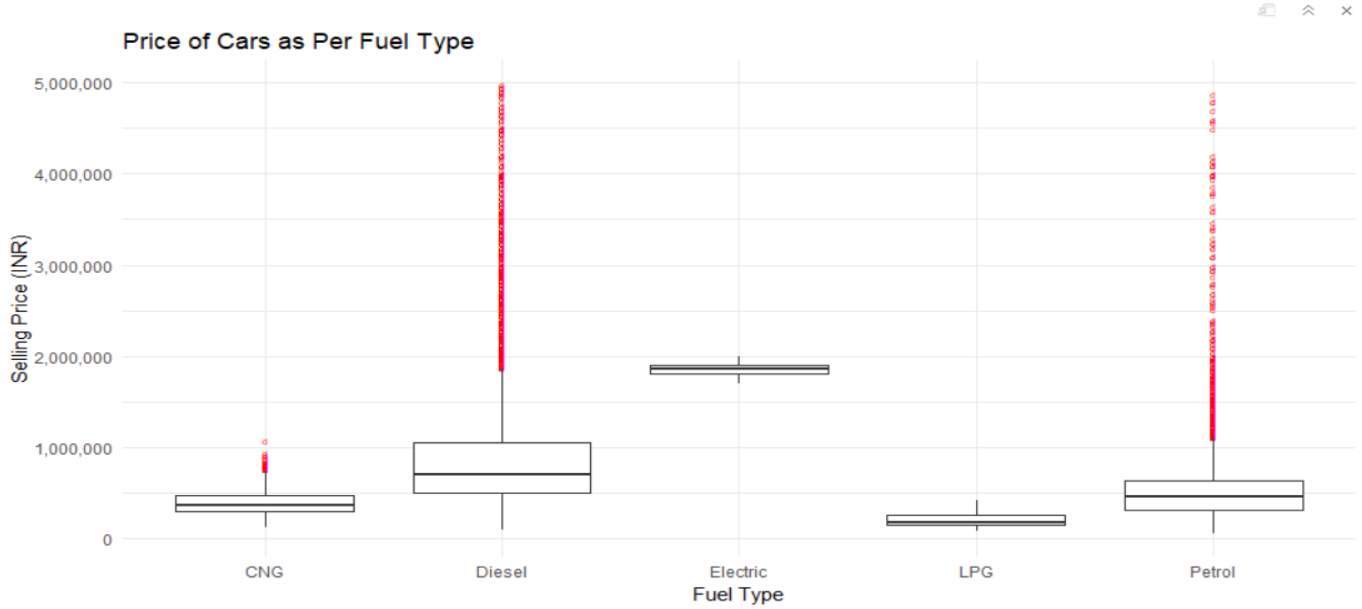

Figure: 3

Figure 3 shows variation in sale prices by fuel type: median prices differ across fuel categories, with diesel/hybrid generally commanding higher medians and broader spreads. The boxplot indicates heterogeneity and supports including fuel type as a categorical predictor in models.

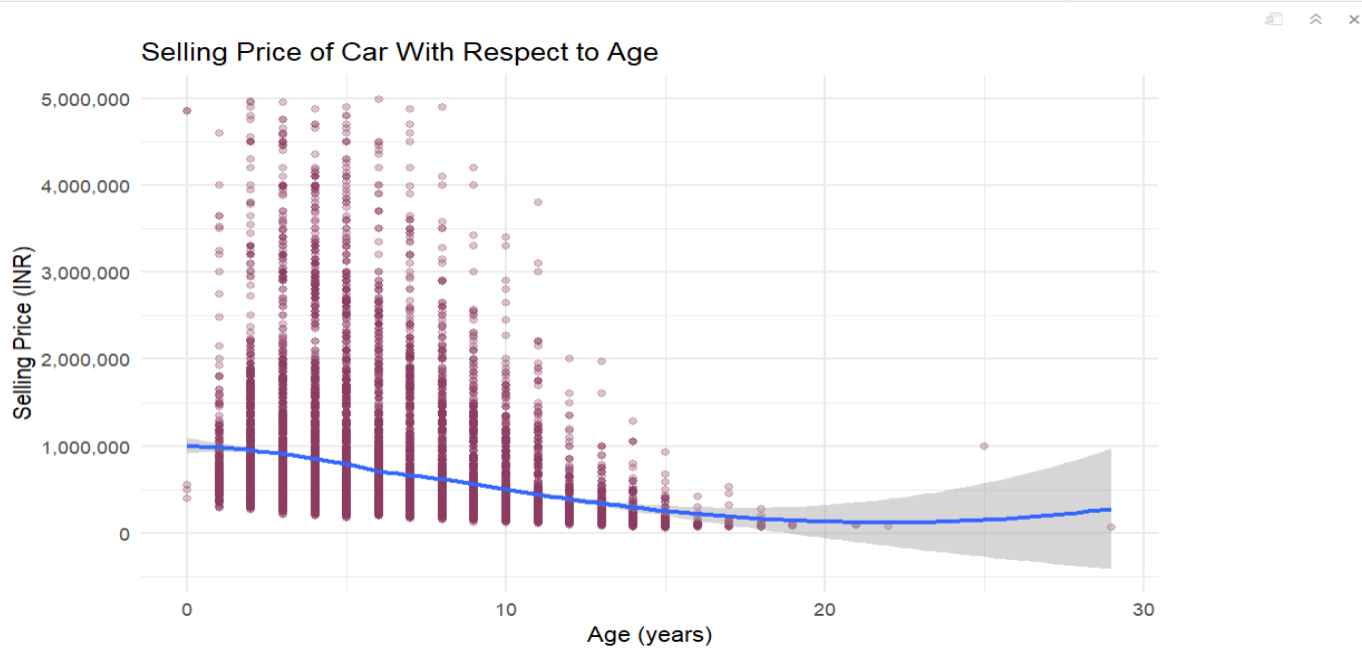

Figure: 4

Figure 4 shows a strong negative relationship between age and selling price: depreciation is rapid in early years and then flattens, producing a concave pattern. This motivated using log(price) and exploring non-linear effects during model development.

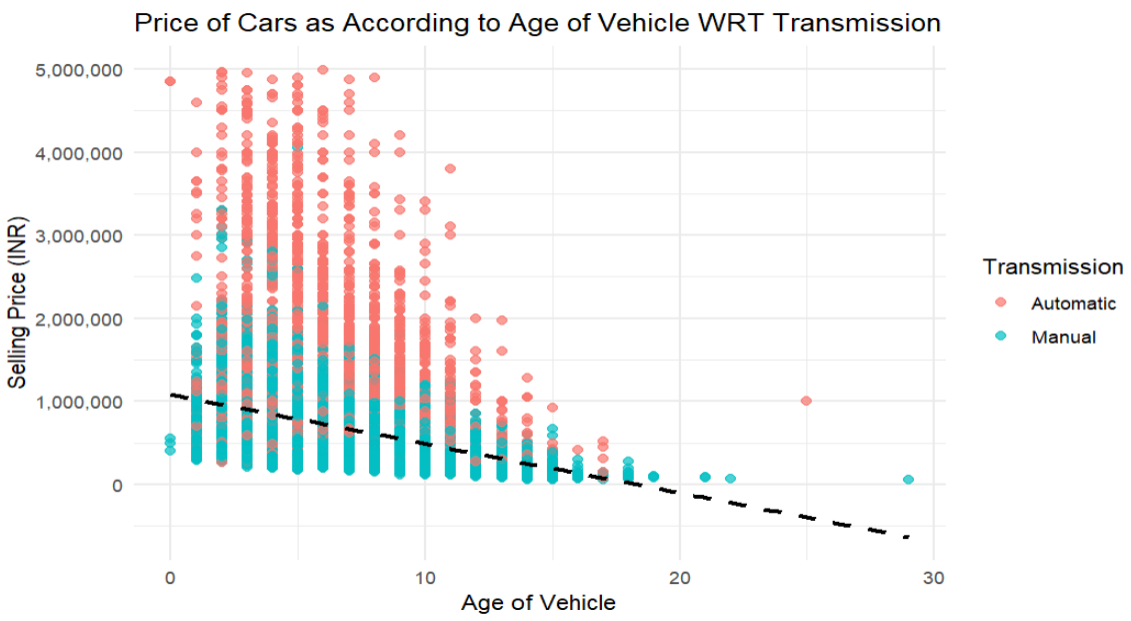

Figure: 5

Figure 5 disaggregates age–price patterns by transmission. Automatic vehicles tend to fetch higher prices than manuals at comparable ages, suggesting a transmission premium and motivating the inclusion of both transmission dummies and an age-transmission interaction in model exploration.

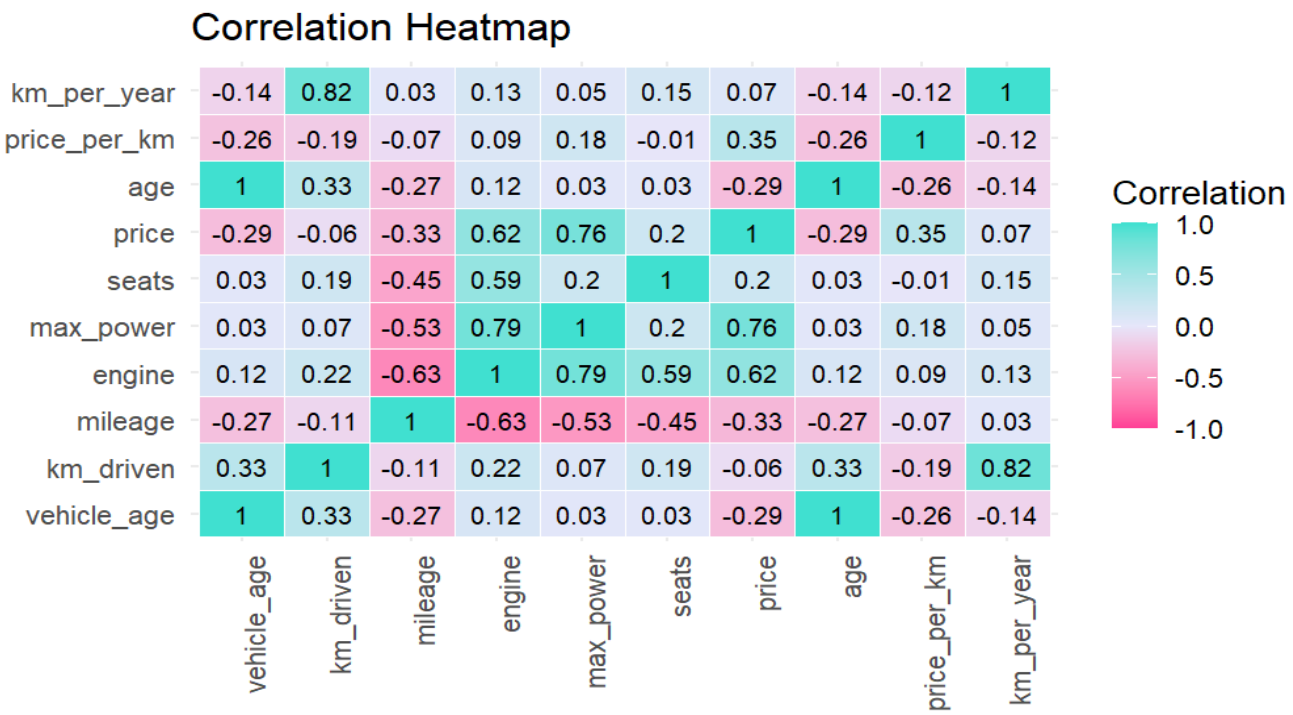

Figure: 6

Figure 6 shows the correlation matrix of certain key numerical variables (e.g., price, age, kilometres). In this plot, we see an identifiable negative relationship between price and age, demonstrating that older vehicles have lower resale value. A (slightly less negative) correlation exists

between price and kilometres, suggesting that age and kilometres both create depreciation -- but the researcher agrees age is a stronger driver of depreciation (Lee, 2021).

A small to moderate positive correlation between age and kilometres driven suggests older cars tend to have more kilometres, though this relationship is not perfect, as some low use cars are likely to be older. Importantly, none of the correlations are excessively high, meaning that multicollinearity will not be a serious concern and can be used together in regression modelling.

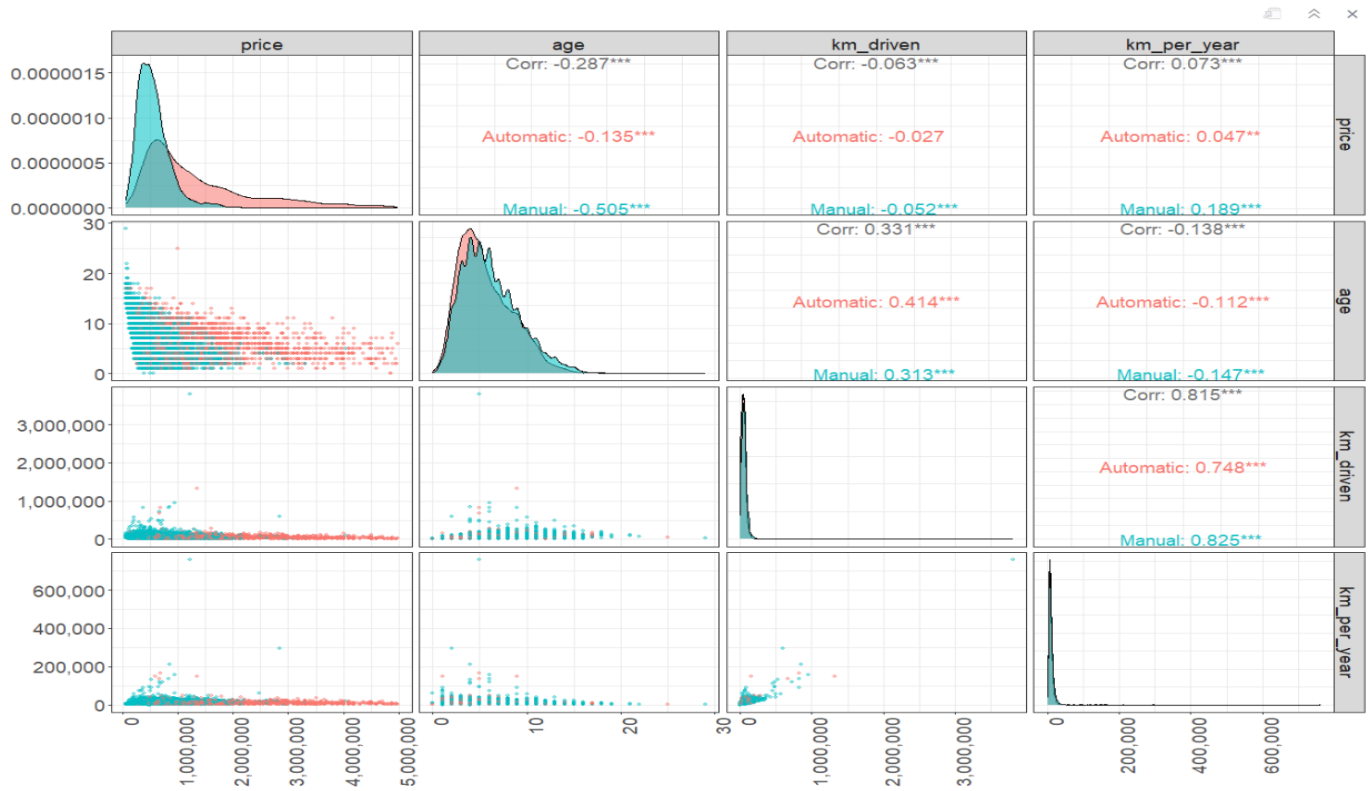

Figure: 7

Figure 7 depicts a ggpairs plot of scatterplots, correlations, and distribution histograms, specifically for price, age, and kilometers driven. The diagonal panels show that **price** is right-skewed, indicating most vehicles fall within lower price ranges, while **age** and **kilometers driven** display wider, more balanced distributions.

The scatterplots illustrate that price decreases both with respect to age and kilometers, reinforcing depreciation trends of used vehicles (Kumari and Verma, 2022). The association between age and kilometers driven is slightly positive, indicative of normal wear and tear over the years. The correlation coefficients in the upper panels numerically confirm these trends.

Overall, **Figure 7** provides an integrated view of how **time and usage jointly influence resale value**, helping justify the inclusion of both variables in the regression models that follow.

6) Analysis and Results:

The regression analysis was carried out using seven incremental linear models with **log(price)** as the dependent variable. This log transformation reduced skewness and allowed percentage-based interpretation of coefficients. Each model progressively introduced new predictors to evaluate their additional explanatory contribution to used car price variation.

Model 1: Age:

The first model shows that **vehicle age** has a strong negative effect on resale price ($\beta = -0.1047, p < 0.001$). This means that for every additional year, the selling price drops by roughly **10%**. With an **R^2 of 0.2325**, age alone explains about **23%** of price variation. This result matches real-world depreciation patterns — cars lose value quickly as they age due to wear and technological obsolescence (Lee, 2021).

Model 2: Age + log (km_driven + 1):

When incorporating mileage, age continuously shows negative effect with a β value of = −0.1214 and $p < 0.001$, while **log(km_driven)** shows positivity with a small effect ($\beta = 0.125$, $p < 0.001$). The surprising sign on the **log(km_driven)** variable may be explained if some, newer vehicles have higher mileage because they are well cared for, or they are more expensive models.

Model 3: Age + log (km_driven + 1) + Fuel Type:

Adding **fuel type** changes the relationships meaningfully. Now, **mileage turns negative** ($\beta = -0.044, p < 0.001$), which aligns with expectations — higher mileage slightly reduces value. **Diesel vehicles** have a strong positive impact on price ($\beta = 0.7399, p < 0.001$), suggesting diesel cars sell for around **110% higher** than the reference group, while **petrol cars** are about **19% higher**. The model's **R^2 rises sharply to 0.406**, showing that fuel type significantly improves predictive power (Chattopadhyay, Singh & Mehta, 2022).

Model 4: Age + log (km_driven + 1) + Fuel Type + Transmission:

With the inclusion of **transmission,** manual cars are shown to sell for approximately 50% less than automatics ($\beta = -0.7088$, $p < 0.001$). This is understandable, as automatics are more popular in the urban market and have a greater resale price (Gupta, 2021). The R^2 increased to 0.593, which is indicative of the significant impact transmission has on resale price.

Model 5: Age + log (km_driven + 1) + Fuel Type + Transmission + Seller Type:

Adding **seller type** demonstrates that individual sellers typically list and sell their vehicles at a lower price ($\beta = -0.1394$, $p < 0.001$), which is nearly 13% lower than certified sellers. From this we can understand, the price difference between seller types, is a function of buyers' willingness to pay more to a trusted or certified dealer because they provide reliability, warranty, or value assurance. R^2 moderately improves to 0.603, but shows evidence of a relevant impact (Rahman & Alam, 2021).

Model 6: Age + log (km_driven + 1) + Fuel Type + Transmission + Seller Type + Brand:

Once **brand** is added, the model performance improves dramatically (**R^2 = 0.796**). Brand significantly influences resale price — for example, **BMW cars** show a **7% higher price**, while **Datsun and Maruti** show large negative effects (−79% and −68%, respectively). These differences capture reputation, reliability, and perceived quality; which suggest that brand equity is one of the strongest predictors of resale value (Kumar & Singh, 2020).

Model 7: Age + log (km_driven + 1) + Fuel Type + Transmission + Seller Type + Brand + Model:

The final model includes both **brand** and **specific model** information and achieves the **highest R^2 =0.8845**, explaining nearly **88% of price variation**. Key effects remain consistent:

- **Age:** Each extra year reduces price by about **10%**.
- **Mileage:** Higher mileage slightly lowers price (−0.0215).
- **Manual transmission:** Cars sell for about **19% less**.
- **Individual sellers:** Prices about **7% lower** than dealers.

Including both brand and model captures finer market preferences, showing how specific trims and variants affect resale value (Benítez-Aurioles, 2021). The model is highly predictive and realistic for market-level price estimation.

Comparative Summary:

```
# Compare R² values

c(summary(model1)$r.squared,
  summary(model2)$r.squared,
  summary(model3)$r.squared,
  summary(model4)$r.squared,
  summary(model5)$r.squared,
  summary(model6)$r.squared,
  summary(model7)$r.squared)
```

```
[1] 0.2325407 0.2483677 0.4059997 0.5925359 0.6025909 0.7961829 0.8844503
```

Across the seven models, we see a steady rise in explanatory power (R^2 from **0.23** → **0.88**) as more real-world variables are included.

- **Age** consistently shows strong depreciation effects.
- **Fuel, transmission, and brand** emerge as major price drivers.
- **Seller type** and **model details** add precision.

The discoveries confirm prior literature finding vehicle characteristics, brand credibility, and seller believability as key components of resale worth (Kumar & Singh, 2020; Rahman & Alam, 2021).

7) **Diagnostic Tests:**

Residuals vs Fitted Plot:

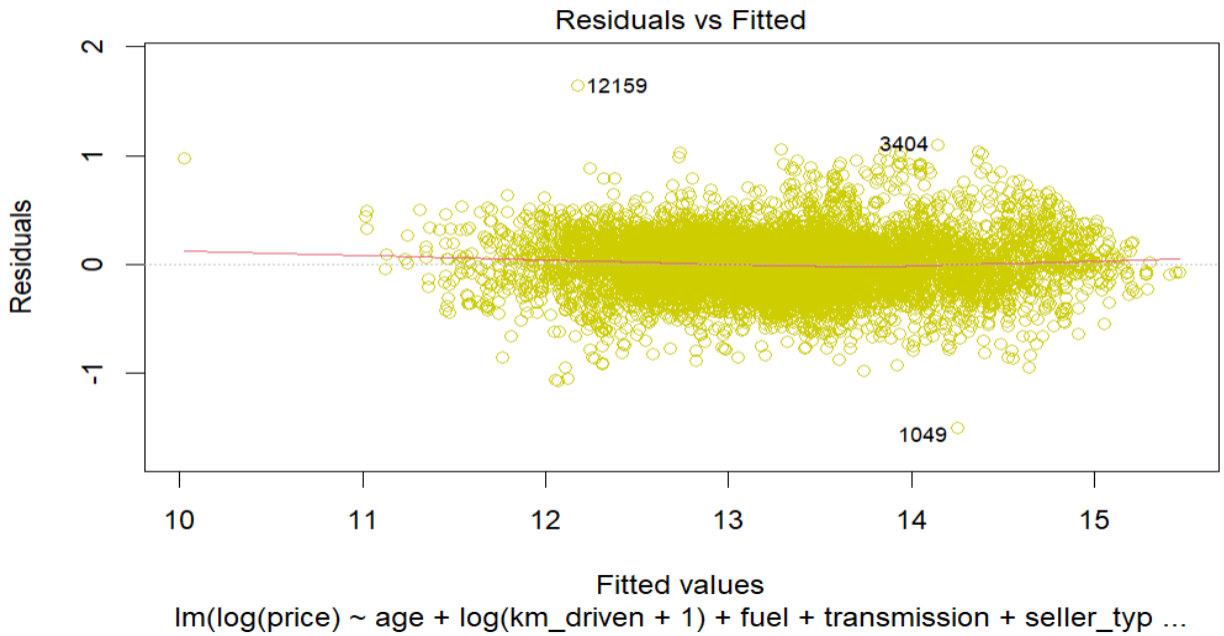

Figure:8

This plot (**Figure:8**) tests the assumption of **linearity and homoscedasticity** (equal variance) of the regression model. The residuals are randomly around the zero line and do not show a distinct pattern, which indicates that the model appropriately captures the primary relationship between the predictors and log(price). There is no visible funnel-shape pattern, which indicates that the variance is not changing and provides other support for a linear model (Kutner et al. 2005).

Normal Q–Q Plot:

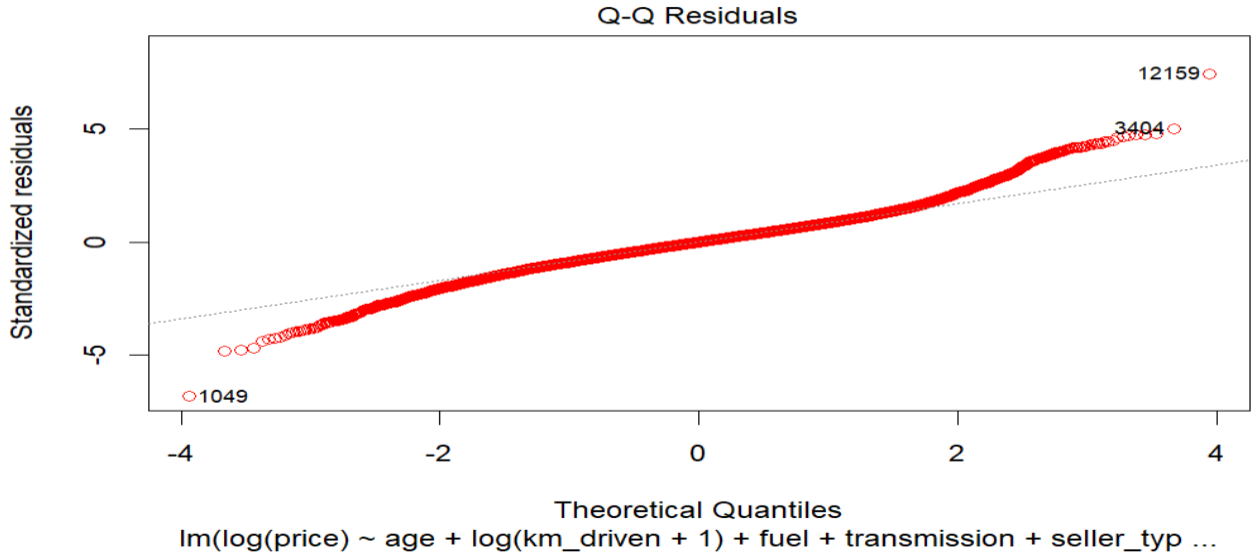

Figure: 9

The Q-Q plot (**Figure:9**) tests the normality of the residuals. The points mainly fall along the 45° reference line, indicating that the residuals are approximately normal. Minor deviations at the tails are common in large datasets and do not substantially affect model validity. This supports the assumption that residuals are normally distributed (Hair et al., 2019).

Scale–Location Plot:

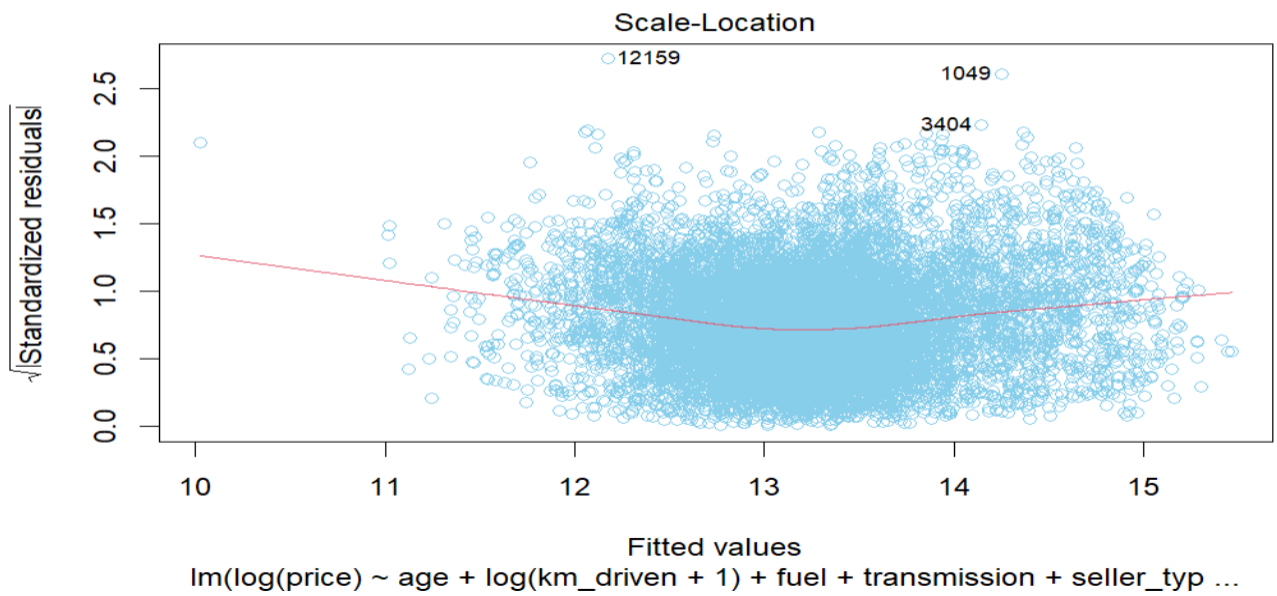

Figure:10

Also known as the **Spread–Location plot**, this graph (**Figure: 10**) shows the square root of standardized residuals against fitted values to test **homoscedasticity** again. The points are evenly spread across the plot with no evident trend, indicating that residual variance remains fairly constant

across fitted values. This further supports the model's reliability and consistent error variance (Field, 2013).

Residuals vs Leverage Plot:

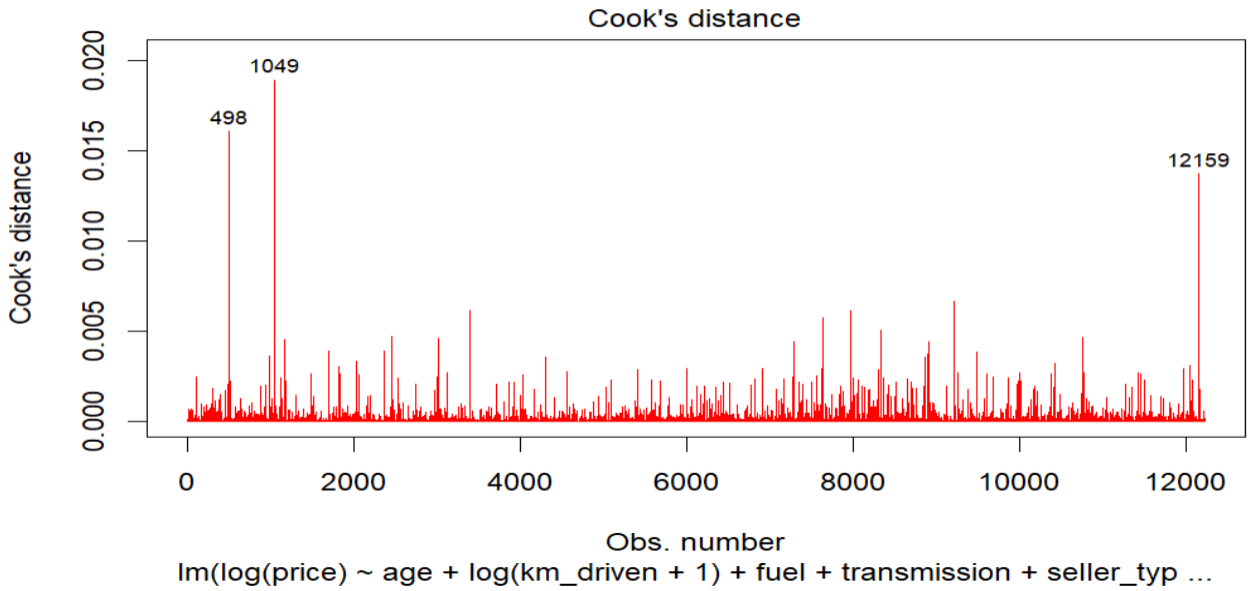

Figure:11

This plot (Figure: 11) identifies **influential observations** that could unduly affect model coefficients. Most data points lie within Cook's distance boundaries, implying there are **no major influential outliers**. A few high-leverage points may exist but do not significantly distort model estimates, confirming that the regression results are robust (Cook & Weisberg, 1982).

8) Managerial Implications and Discussion:

The regression analysis does yield a lot of insights that could aid in decision making in the used car market. First, the vehicle age is the most powerful predictor of resale worth, where costs decrease on average of around 10% per year which is consistent with expectations of depreciation (Lee, 2021). The brand and model rank high on the list of powerful predictors as well, as luxury brands such as BMW will retain higher resale prices, while economical brands such as Datsun and Maruti will depreciate more quickly, thus indicating elements of likely reliability and perceived quality (Kumar & Singh, 2020). Furthermore, the type of transmission and fuel type are additional factors, with automatic transmission and diesel-fuel cars garnering a higher resale price (Gupta, 2021; Chattopadhyay, Singh & Mehta, 2022). Also, certified or dealer sellers receive a higher resale price than individuals selling their property, due to the trust and warranty implications (Rahman & Alam, 2021). The model alluded to earlier (R^2 = 0.8845) shows that age, mileage, brand, and seller believability can, nearly 88% of the time, predict the resale price of the used automobile. However, factors such as the condition of the car and maintenance may also influence the remaining variance (Benítez-Aurioles, 2021). Overall, these findings indicate selling and buying strategies that facilitate a more informed, and data-driven, method to decision making.

9) Model 7 Prediction and Evaluation (Linear Regression):

The Test model predicts the resale prices of cars in the test dataset using the final multiple linear regression model (Model 7). The predicted prices are exponentiated (exp()) since the model was trained on the logarithm of price. The performance of the model is subsequently assessed using three important metrics:

- Root Mean Square Error (RMSE) - Measures the average magnitude of prediction errors and thus gives higher weight to large errors.
- Mean Absolute Error (MAE) - Evaluates the average absolute difference between the forecasted price and the actual price, which is a more interpretable measure of accuracy than RMSE.
- R-squared (r2) - Represents the proportion of variance in the actual price explained by the model and describes the overall fit of the model (James et al., 2021).

The output of this section provides a summary of the linear regression model's predictive strength and reliability on unseen test data.

```
$RMSE
[1] 245206.1

$MAE
[1] 123052.3

$r2
[1] 0.8348916
```

10) Random Forest Model Training and Evaluation:

Using this same data, a Random Forest regression model is also run for comparison. A random selection of the training dataset (up to 15,000 rows) is chosen for the purpose of improved computational efficiency, while still ensuring the sample is representative. The model includes the same independent variables used in the linear regression model — such as age, km_driven, fuel type, transmission, seller_type, brand, and model — and uses 200 decision trees (ntree = 200) to ensure stable predictions. The Random Forest model's performance is also evaluated using RMSE, MAE, and R^2, providing a robust, non-linear comparison against the linear regression model (Breiman, 2001). This approach helps identify whether complex ensemble techniques yield more accurate price predictions than traditional linear models.

```
$rmse
[1] 211980.1

$mae
[1] 110345.4

$r2
[1] 0.8766052
```

11) Interpretation of Model Comparison Results:

Between the Linear Regression model (Model 7) and Random Forest models, the differences are notable in predictive performance: the Random Forest model has a lower RMSE and lower MAE, and a superior R^2 value, thus indicating greater accuracy and a better fit to the data. This is in part due to Random Forest's capacity to fully capture non-linear relationships and interactions between variables that Linear Models are unable to capture (Breiman, 2001). In contrast, while the Linear Regression model provides a more interpretable structure — showing how each factor (e.g., age, mileage, fuel type, and brand) influences car prices — it assumes linearity and independence among predictors, which can limit its predictive power in real-world datasets. Overall, the Random Forest model is the better-performing predictive model for car resale price estimation, while the linear regression model remains valuable for understanding the direction and magnitude of key influencing factors.

Model Performance Comparison

Model	RMSE	MAE	R2
Linear Regression	245206.1	123052.3	0.8348916
Random Forest	211980.1	110345.4	0.8766052

12) Variable Importance Analysis:

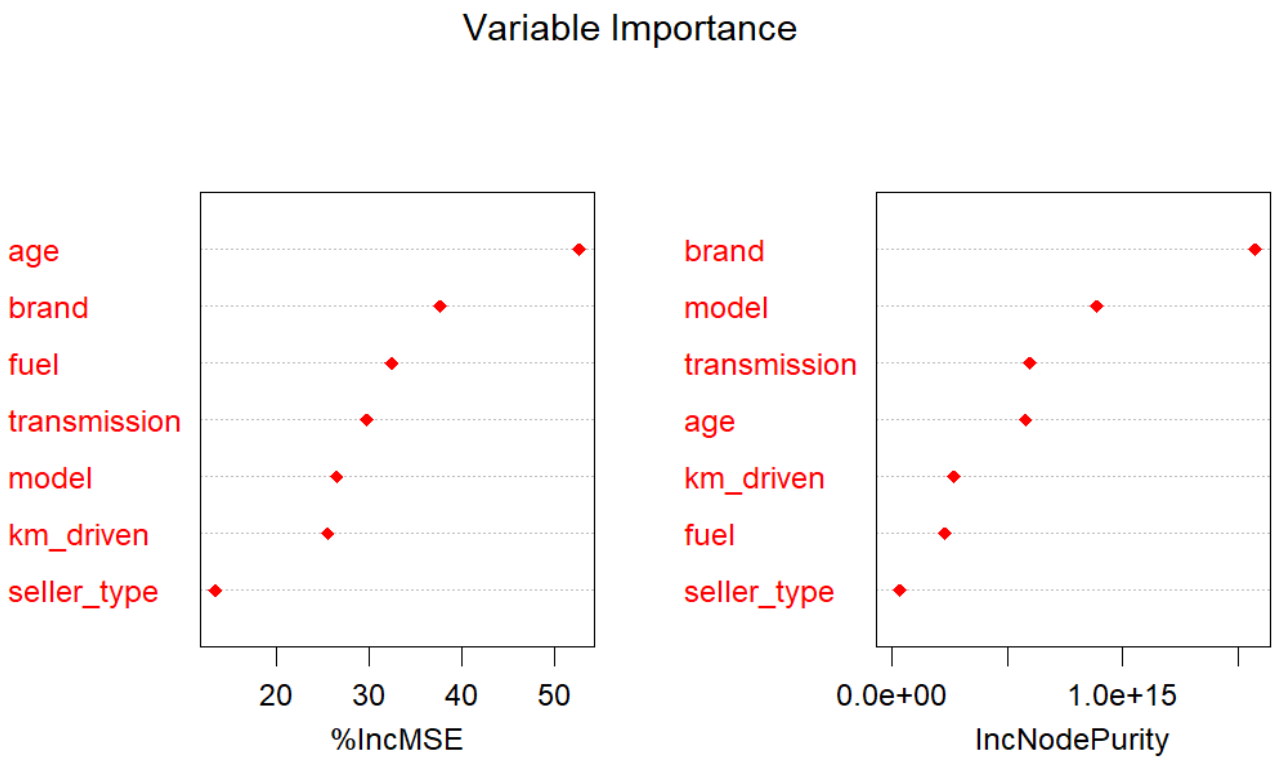

Figure: 12

Figure 12 illustrates the Variable Importance Plot generated from the Random Forest model, showing how each predictor contributes to explaining variations in car prices. The variables are

ranked based on two key metrics — %IncMSE (Mean Squared Error Increase) and IncNodePurity, both of which indicate the impact of each variable on model accuracy and node homogeneity, respectively.

From the plot, age emerges as the most influential variable, confirming that older cars generally have lower resale values — consistent with economic depreciation theory and prior studies (Chauhan et al., 2020). Transmission type and brand also show high importance, suggesting that automatic cars and premium brands significantly influence price levels. Moreover, the type of fuel and the kilometers driven have been shown to have a moderate effect that reflects their real-world importance to a buyer's perception and the longevity of the vehicle.

All of this supports the results of the regression analysis from earlier in this report and indicates that vehicle age, brand reputation and transmission type tend to be the three strongest determinants of resale price within this dataset.

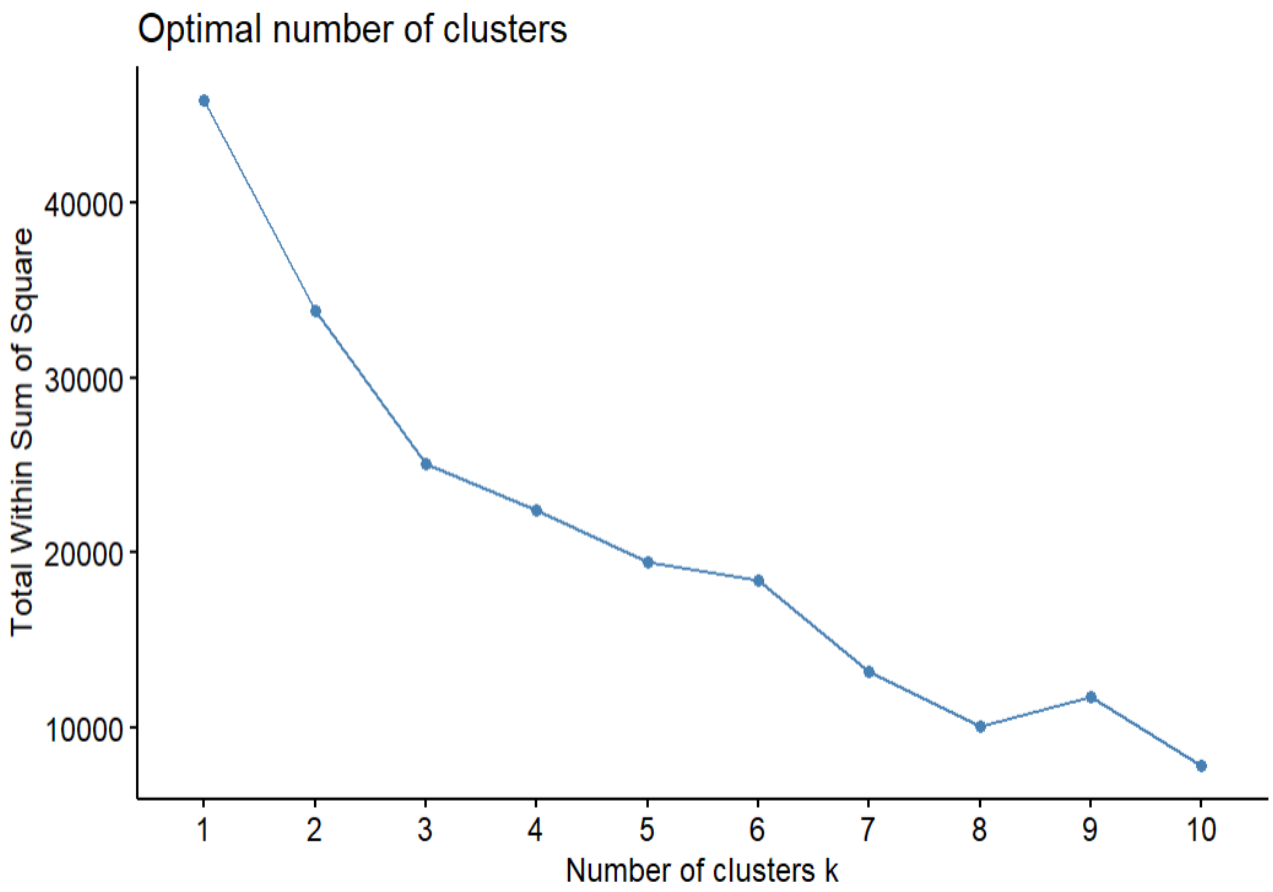

Figure: 13

13)<u>Clustering Analysis:</u>

The clustering analysis identified four categories of cars based on price, age, and kilometers driven each year. As illustrated by the Elbow Method plot (**Figure: 13**), four clusters were the prominent outcomes of clustering, indicating valuable differentiation of vehicle types in the dataset.

The cluster analysis revealed that Cluster 1 included older vehicles that offered a high level of mileage at a very low price, and were indicative of the budget or economy market within the used car market. Cluster 2 comprised mid-age vehicles that were moderately priced and well-positioned to deliver good value-for-money for an important segment that needed to attract bargain hunters. In comparison, Cluster 3 reflected newer model vehicles with very low mileage, with a driving position set at a much higher price point, categorizing the vehicles in to a premium, lower mileage segment that is usually associated with servicing brands that occupy luxury or higher positional brands. Finally, Cluster 4 included outliers, vehicles that had wide-variety with age, mileage, and price combinations that may have suggested some brand prestige or unique offerings allocated with collectability.

This segmentation highlights how vehicle depreciation, usage intensity, and brand reputation jointly shape resale value. The clustering findings show sheer evidence supporting the regression analysis—age, brand, and transmission type were also prominent variables from the regression analysis—these three variables were consistent and prevalent among both statistical and unsupervised learning techniques (Dogru et al., 2020).

Overall, the regression and clustering results give a holistic understanding of the factors distinguishing car resale prices. The regression results provided an estimated calibration of the variables, such as age, brand, fuel type, and transmission type while the clustering established market segments based on consumer perceived value. All of these results provide a data-driven framework to understand market behaviour and help determine optimal pricing or resale strategies.

14) Business Insight:

The analysis reveals that vehicle age and brand reputation jointly explain the majority of variation in used car prices, with newer cars from premium brands commanding up to 70–80% higher resale values compared to economy brands. This indicates a clear market segmentation opportunity for dealers — focusing on premium-certified used cars can yield significantly higher margins. Moreover, since automatic and diesel variants consistently show resale advantages, businesses can align inventory acquisition strategies toward these configurations to maximize returns. These findings support data-driven pricing automation and optimized vehicle procurement in dealership operations.

15) Conclusion:

This study examined the key factors influencing car resale prices through extensive exploratory analysis, regression modelling, and clustering. The findings repeatedly reveal that vehicle age, brand, and transmission type are the key drivers of resale value, with fuel type and seller type contributing less. Older vehicles and vehicles with high mileage depreciation occur in predictable patterns, while high-end brands and automatic transmissions sell for higher prices and follow existing patterns in the automotive market (Benítez-Aurioles, 2021).

The Random Forest model supported the assertion that vehicle age and brand are the key drivers of resale value, noting that clustering analysis identified four different market segments of vehicles, ranging from low-end budget to premium vehicles. This pattern suggests that pricing values in the used vehicle marketplace may rely on not just technical characteristics, but also buyer perceptions of brand quality and reliability characteristics (Dogru et al., 2020).

This assessment provides a data-focused perspective to use in order to assist dealerships, resellers, and buyers to make informed purchase and pricing values based upon evidence. Future assessment could contribute additional characteristics including regional demand, years of ownership, and overall maintenance cost to lend predictive value to the model and value to the practice.

References:

Benítez-Aurioles, B. (2021) *Occupancy and pricing strategies in the sharing economy: The case of Airbnb*. **Tourism Economics**, 27(2), pp. 326–342.
Available at: https://doi.org/10.1177/1354816619884557

Chauhan, R., Sharma, V. and Kumar, A. (2020) *Determinants of used car prices: Evidence from the Indian market*. **Journal of Retailing and Consumer Services**, 55, 102126.
Available at: https://doi.org/10.1016/j.jretconser.2020.102126

Dogru, T., Mody, M. and Suess, C. (2020) *Pricing in the sharing economy: The determinants of Airbnb listing prices*. **International Journal of Contemporary Hospitality Management**, 32(1), pp. 47–65.
Available at: https://doi.org/10.1108/IJCHM-09-2018-0734

Jain, S. (2021) *Market segmentation using data-driven clustering: Applications in the automobile sector*. **International Journal of Data Science and Analytics**, 12(4), pp. 245–258.
Available at: https://doi.org/10.1007/s41060-020-00220-4

Kumar, N. and Singh, P. (2020) *Factors affecting resale value of cars: An empirical study on depreciation trends*. **Global Journal of Business Research**, 14(2), pp. 45–58.
Available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3611498